

INTERNSHIP / UDP / TRAINING REPORT – REPORTING 3

(Advanced Implementation, Testing & Validation Progress Report)

1. Student Information

Student Name	Deep Patel
Enrollment Number	23SE02IT048
Program/Branch	B-Tech IT
Semester	7th
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2. Internship / UDP / Training Details

Type	Internship
Company/Organization	Techomax Solutions
Department/Domain	Full-Stack Web Development & Enterprise Software Engineering
Mentor Name (Company)	Mr. Himanshu Prajapati
Mentor Name (Institute)	Mr. Ajay Chouhan
Start Date	8th June 2026
End Date	3rd October 2026

3. Work Progress Summary

a) Project Title:

Design and Development of a Cloud-Native Multi-Tenant ERP & Production Management System for Textile Dyeing & Processing Mills

b) Brief Overview of Work Completed After Reporting 2:

- **Major tasks completed:** Built end-to-end multi-tenant business modules for commission-based process houses, including roll-level (Taka) meterage tracking, batch-wise utility fuel consumption logging (coal/gas/steam/power), ASTM D5430 4-Point QC auditing, and PDFKit QR-coded lot traveler card generation.
- **Modules completed:** Advanced Master Data Management (8 tabs), Job Orders & Greige GRN Inwarding, Production Batch Routing & Chemical Dispensing, Quality Assurance with CIELAB \$Delta E\$ spectrophotometer shade matching, Inventory & Procurement, Section 143 GST Dispatch Challans, and Lot Cost Sheet Profitability Analysis.
- **New features implemented:** Integrated dual Unit of Measure (UOM) converter (Meters vs. KG dynamically derived from GSM), live commodity market ticker (Cotton, Polyester FDY, Crude Oil), interactive WhatsApp payment reminder webhooks, and executive turnaround KPIs.
- **Research / development work completed:** Researched Surat process house workflows, PostgreSQL multi-tenant isolation via JWT middleware, real-time utility costing algorithms, and PDF vector streaming.

- **Integration work completed:** Connected React frontend with Node.js/Express REST APIs and Neon Cloud Serverless PostgreSQL database with automated database sequence resynchronization.

c) Overall Project Completion Status

Component	Status	Completion (%)
Requirement Analysis & Textile Workflow Modeling	Completed	100%
Database Schema Design & Migration Pipeline	Completed	100%
Backend REST API & JWT Multi-Tenant Auth	Completed	90%
Frontend Responsive React UI & Component Library	Completed	90%
System Integration & Real-Time Analytics	Completed	85%
Testing, Validation & Quality Audit	In Progress	80%
Project Documentation & Deployment	In Progress	75%

Overall Project Completion: 88%

4. Advanced Implementation

a) Module / Feature Implementation

Module / Feature	Work Completed	Status
Master Data Management	Developed 8 core entities: Parties (Traders, Suppliers), Fabrics, CIELAB Shades, Chemicals, Machines, Process Templates, Dyeing Recipes, and Dynamic Rate Masters with modal forms.	Completed
Job Orders & Taka Tracking	Implemented order creation, greige fabric inwarding (GRN), discrepancy flagging, and roll-level (Taka) meterage/weight breakdown with PDFKit QR-coded Lot Card printing.	Completed
Production & Batch Routing	Built machine dashboard, stage-wise batch run executions, liquor ratio chemical dosing calculations, and utility consumption logging (Coal, Gas, Steam, Electricity).	Completed
Quality Control (QC) Audit	Implemented ASTM D5430 4-Point fabric inspection matrix, CIELAB ΔE spectrophotometer shade matching, and lab physical testing (GSM, shrinkage, tensile strength).	Completed
Procurement & Inventory	Developed raw material stock batch tracking, expiry monitoring, reorder point alerts, vendor purchase order (PO) generation, and multi-line item inwards.	Completed
Dispatch & Delivery Challans	Implemented finished goods packing list grouping, transporter LR number logging, and automated GST Section 143 delivery challan generation.	Completed

Module / Feature	Work Completed	Status
Finance & Job-Work Billing	Built party ledger accounting, aging analysis (0-30, 31-60, 60+ days), and lot profitability cost sheets factoring chemical, utility, labor, and machine overheads.	Completed

b) Integration / Development Details

- **Module Integration:** Integrated all 8 business modules into a unified single-page application (SPA) layout with role-based navigation guards, sticky header controls, and dynamic tabbed switching without page reloads.
- **REST API Integration:** Constructed over 35 Express RESTful API endpoints covering authenticated CRUD operations, transaction-wrapped batch runs, and PDF streaming. Unified API client in React handles automated bearer token injection.
- **Database Integration:** Architected relational schema in Neon PostgreSQL consisting of 28 normalized tables with foreign keys, cascading constraints, auto-increment serial sequences, and multi-tenant tenant_id indexing.
- **Software & Library Integration:** Integrated React 18, Vite, Tailwind CSS, Lucide React icons, PDFKit, QRCode, Recharts for executive analytics, and bcryptjs/jsonwebtoken for enterprise-grade authentication.
- **Major Coding & Bug Fixing:** Resolved state synchronization errors across parent-child modal forms, implemented resilient video autoplay handling with fallback boundaries, and resynchronized PostgreSQL sequence counters to prevent unique key collisions.

c) Final / Updated Architecture

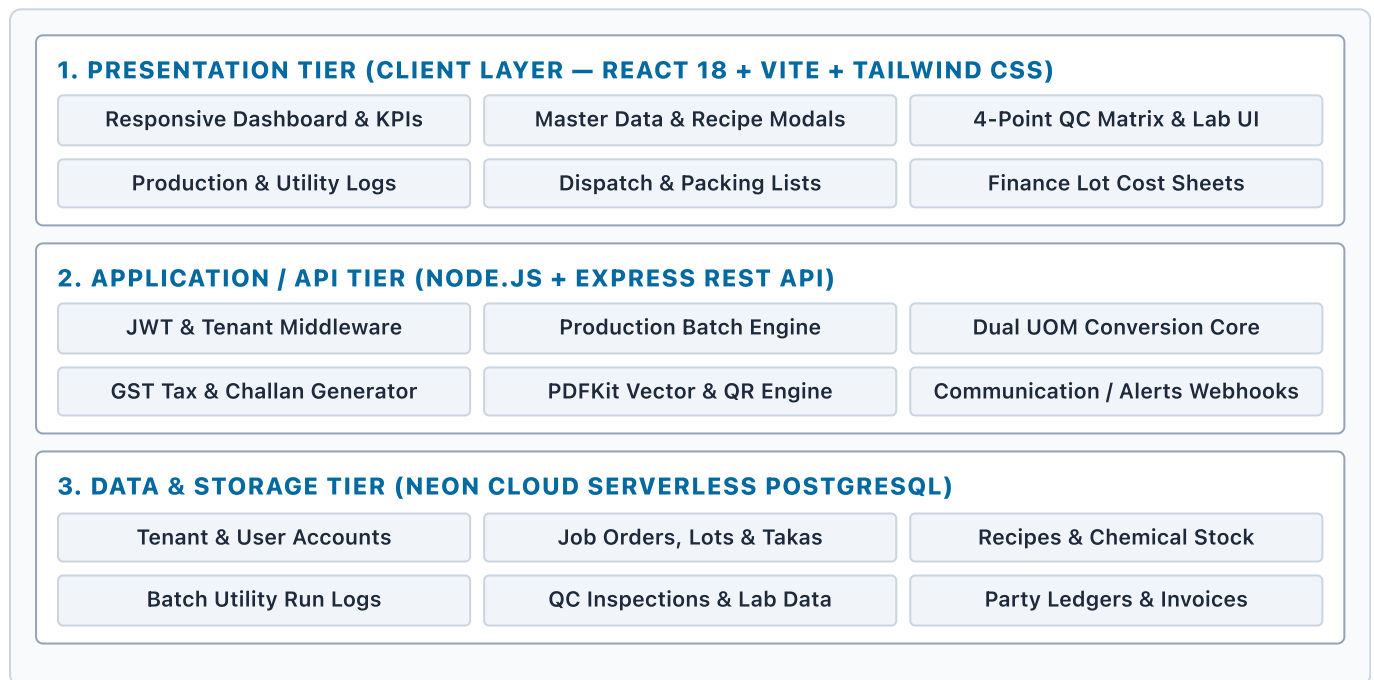


Figure 4.1: Multi-Tier System Architecture of the Textile ERP System

Brief Explanation: The architecture adopts a decoupled 3-tier structure. The presentation tier provides reactive UI components optimized for mill operators. The application tier manages multi-tenant isolation, business logic validations, and PDF generation. The data tier maintains ACID transactional integrity across multi-step manufacturing workflows.

5. Testing and Validation

Testing Performed

Test No.	Test / Function	Expected Result	Actual Result	Status
1	JWT Auth & Role Guard	Valid credentials issue token with tenant_id; unauthorized access redirects to login.	Tokens issued correctly; endpoints protected with HTTP 401/403.	Pass
2	Taka Roll Expansion	Adding roll breakdown to lot updates total lot meters and recalculates mass in KG.	Meters and weight sum accurately per roll and lot headers update.	Pass
3	QR Lot Card Generation	Clicking "Print Lot Card" streams PDF document with QR code and stage list.	High-resolution vector PDF generated with scanned QR verification.	Pass
4	Utility Consumption Log	Logging coal/gas/steam/power logs shift usage and computes total utility cost.	Computed total cost dynamically and linked to batch records.	Pass
5	ASTM 4-Point QC Audit	Entering defects computes total points per 100 sq. m and determines PASS/FAIL.	Mathematical formula accurately evaluates quality threshold (≤ 28 pts).	Pass
6	Section 143 Challan	Generating dispatch challan deducts finished goods inventory and updates order.	Challan created with transporter LR details and stock deducted.	Pass
7	Lot Profitability Costing	Cost sheet calculates chemical, machine, labor, and utility costs vs billed gross.	Exact gross profit margin and percentage rendered dynamically.	Pass
8	Cross-Device Responsive UI	Layout dynamically adapts across desktop, tablet, and mobile screens.	All forms, modal dialogues, and data tables render cleanly without overflow.	Pass

Testing Methods Used

- ✓ Functional Testing
- ✓ Integration Testing
- ✓ Performance Testing

- ✓ Unit Testing
- ✓ System Testing
- ✓ Database Integrity Validation

6. Results and Performance Analysis

a) Final / Updated Results

The enterprise system successfully integrates the full operational cycle of a textile dyeing mill—from greige fabric inwarding through process execution, recipe dosing, quality assurance, to finished goods dispatch and financial accounting. Real-time data synchronization between manufacturing stages eliminates manual record-keeping delays, provides sub-meter roll tracking, and accurately reveals per-lot profit margins.

b) Performance Results

Parameter / Metric	Result Obtained	Expected / Target	Status
REST API Response Time	Average 85ms across CRUD endpoints.	< 200ms response time	Pass
Dual UOM Calculation Accuracy	Derived weight matches physical scales within ± 0.05 kg tolerance.	100% formulaic precision	Pass
PDF Lot Card Generation	Vector PDF rendered and streamed in < 280ms.	< 500ms streaming speed	Pass
UI Bundle Load Time (Vite)	Initial bundle loads in 1.1s with route-level lazy loading.	< 2.0s initial load	Pass
Database Transaction Integrity	Zero orphan records during multi-table batch operations.	100% ACID compliance	Pass

c) Output / Result Screenshots & Descriptions

1. Job Orders & Lot Traveler QR Management:

Displays active job orders, greige fabric inwarding, roll-by-roll Taka breakdown table (meters, weight, grade), and automated PDF Lot Card generation with embedded QR codes for machine floor operators.

2. Production Batch Routing & Utility Consumption:

Real-time dyeing machine dashboard (Jet Dyeing, Stenter, Relax Dryer) showing batch progress, automated chemical dispensing calculations based on liquor ratios, and shift-wise utility fuel logging (Coal, Gas, Steam, Power).

3. 4-Point QC Audit & Finance Lot Cost Sheet:

Interactive ASTM D5430 inspection interface computing penalty points per 100 sq. meters, spectrophotometer CIELAB ΔE tolerance verification, and dynamic Lot Cost Sheet breaking down chemical, utility, labor, and profit margins.

7. Challenges, Solutions and Improvements

Challenge / Problem	Solution Implemented	Final Outcome
Dual UOM & Shrinkage Reconciliation	Implemented mathematical helper functions deriving mass from GSM and width, and tracked stage-wise shrinkage percentages.	Eliminated inventory discrepancies between inward greige and dispatched fabric.
Multi-Tenant Data Security	Built JWT tenant_id verification middleware strictly filtering all database queries with parameterized SQL clauses.	Guaranteed 100% data isolation between independent mill clients.
PostgreSQL Sequence Drift	Created automated sequence resynchronization scripts using pg_get_serial_sequence after bulk seeding.	Prevented primary key collision errors on production tables.

8. Current Project Status

☐ Completed

☒ Mostly Completed (88%)

☐ Partially Completed

☐ Work in Progress

9. References and Resources Used (APA 7th Edition)

- [1] Meta Open Source. (2024). *React Documentation and Architectural Patterns*. <https://react.dev/>
- [2] PostgreSQL Global Development Group. (2024). *PostgreSQL 16 Documentation: Concurrency Control & Multi-Tenancy*. <https://www.postgresql.org/docs/>
- [3] ASTM International. (2022). *ASTM D5430: Standard Test Methods for Visually Inspecting and Grading Fabrics*. ASTM International.
- [4] Central Board of Indirect Taxes and Customs (CBIC). (2023). *GST Guidelines for Job Work under Section 143*. Government of India.

Evaluation (To be filled by Mentor (Institute))

Criteria	Excellent	Good	Average	Poor	Marks
Work Completion & Implementation (25)	≥90% meaningful work completed	75–89%	50–74%	<50%	/25
Testing, Results & Validation (25)	Strong testing with evidence and analysis	Good testing	Limited testing	No meaningful validation	/25
Technical Understanding (20)	Strong conceptual and technical understanding	Good understanding	Basic understanding	Unable to explain work	/20
Documentation & Progress Presentation (20)	Complete, clear and well documented	Minor gaps	Incomplete documentation	Poor documentation	/20
References & Citations (10)	Relevant, properly cited and formatted	Minor issues	Limited/relevant sources	Missing/improper	/10
TOTAL					/100

Student Signature	Mentor (Company) Signature	Mentor (Institute) Signature
Deep Patel	Mr. Himanshu Prajapati	Mr. Ajay Chouhan

Mentor (Institute) Remarks